G A M E D E S I G N D O C U M E N T

Loreforge

AI-Powered Interactive Fiction Engine

Build Your Own Adventure — Every Story is Unique

Version 2.0 | March 2026 | **CONFIDENTIAL**

Cross-Platform (iOS, Android, PC/Mac) | Flutter + Flame | Multi-Provider AI

1. Executive Summary

Loreforge is an AI-powered interactive fiction platform that combines the narrative depth of visual novels with the player agency of tabletop RPGs and the exploration mechanics of classic text adventures like Zork. Every story is dynamically generated by a multi-agent AI pipeline, ensuring no two playthroughs are alike.

Players configure their experience through a "Session Zero" wizard — selecting language, genre/subgenre (30+ options across fiction and non-fiction), RPG depth (from pure story to full tabletop simulation), tone, protagonist, and plot twist preferences. They can also provide favorite stories (books, TV shows, movies, games) so the AI calibrates narrative style, pacing, and thematic complexity to match their taste.

The engine uses a model-agnostic, multi-provider AI architecture that routes different agent tasks to the cheapest capable model (DeepSeek, Gemini Flash, GPT Nano for high-volume tasks; Claude Haiku or GPT-5 Mini for premium subscribers), with streaming responses so narrative text appears word-by-word for a smooth, immersive UX.

1.1 Core Value Proposition

- Infinite replayability — AI-generated stories mean every playthrough is unique
- Genre mastery — 30+ genres/subgenres, each with codified craft rules that make stories genuinely engaging
- Taste-adaptive — users provide favorite stories and the AI calibrates to match their narrative preferences
- Scalable depth — from pure interactive fiction to full D&D-style RPG with granular toggle control
- Cross-platform with streaming AI — text appears as it's generated, no waiting
- Multilingual — 9+ languages supported natively

1.2 Target Audience

- Primary: Visual novel and interactive fiction fans (18–35) who want more agency and replayability
- Secondary: TTRPG players who want a solo, narrative-driven RPG experience
- Tertiary: Casual readers/gamers curious about interactive AI storytelling
- Quaternary: Non-English speakers underserved by current interactive fiction offerings

2. Session Zero — Configuration & Personalization

Session Zero is the onboarding wizard where players shape their experience before any story begins. It's designed as a premium, immersive card-based flow — not a boring form. Each step uses atmospheric visuals and audio previews.

2.1 Configuration Steps

Step	Category	Details
1	Language	English, Español, Français, Português, Deutsch, Italiano, 日本語, 한국어, 中文 — all AI output generated in selected language
2	Experience Mode	RPG Adventure (stats, dice, inventory, skill checks) or Pure Story (no game mechanics, purely narrative)
3	Setup Method	"Guide me step by step" (full questionnaire) or "I'll describe what I want" (free-text prompt describing ideal story)
4	Genre & Subgenre	30+ genres across fiction and non-fiction (see Section 3). Primary genre + optional subgenre + optional crossover tag
5	Favorite Stories	User names 1–5 books, movies, TV shows, or games. A Reference Agent analyzes these to extract taste profile (themes, pacing, character dynamics, tone) injected into the World Bible
6	Tone / Mood	Dark & Gritty, Epic & Grand, Comedic, Philosophical, Suspenseful Thriller, Melancholic
7	Protagonist (RPG)	Reluctant Hero, Cunning Rogue, Scholar, Warrior, or Custom — each with preset stat distributions
8	RPG Depth (RPG only)	Presets (Story+, Lite RPG, Classic RPG, Hardcore) plus granular toggles for individual systems
9	Plot Twists	ON/OFF toggle. When ON, AI plants foreshadowing seeds and triggers genre-appropriate twists every 4–6 scenes
10	Difficulty (RPG)	Story Mode (forgiving DCs), Classic (balanced), Hardcore (permadeath, high DCs, resource scarcity)

2.2 Favorite Stories — Taste Profile System

When a user provides favorite stories, a one-time Reference Agent call analyzes them and generates a structured taste profile injected into the World Bible for all subsequent AI calls:

- Identified themes (moral ambiguity, power corruption, revenge, found family, redemption, survival)
- Narrative patterns (slow burn to explosive climax, constant escalation, ensemble dynamics, dual timeline)
- Tone markers (dark humor, atmospheric, cerebral, visceral, whimsical, cinematic)
- Character dynamics (antiheroes, complex villains, unreliable narrators, mentor figures, ensemble casts)
- Pacing preferences (methodical investigation, relentless action, contemplative, episodic)

Example: A user who lists "Breaking Bad, The Witcher 3, Gone Girl" gets a taste profile emphasizing moral decay, complex antiheroes, unreliable perspectives, slow-burn escalation, and dark/cerebral tone. Someone

listing "Studio Ghibli, Stardew Valley, The Princess Bride" gets warmth, wonder, gentle humor, found-family bonds, and lighthearted adventure. Both could select "Fantasy" but get fundamentally different stories.

2.3 RPG Depth — Presets & Granular Toggles

Players first choose a preset, then optionally customize individual systems via an "Advanced" panel:

Preset	Stats	Dice/Skill Checks	Inventory	Companions	Crafting	Resource Mgmt
Story+ (choices only)	Hidden	None	Key items only	Narrative only	No	No
Lite RPG	Visible, 3 stats	Simple pass/fail	Basic	1 companion	No	No
Classic RPG	Full 6 stats	d20 + modifiers	Full system	Up to 3	Basic	Light
Hardcore	Full 6 stats	d20, high DCs	Full + weight	Up to 4, can die	Full	Survival tracking

Granular toggles available under each preset: Stats (hidden/simplified/full), Skill Checks (none/simple/d20), Inventory (none/key items/full/weight), Companions (none/1/party), Crafting (off/basic/full), Resource Management (off/light/survival), Permadeath (on/off), Relationship Tracker (on/off), Map System (on/off).

3. Genre Taxonomy & Effectiveness Framework

Loreforge supports 30+ genres organized into Fiction and Non-Fiction Inspired categories. Each genre has codified craft rules that the AI agents must follow to produce stories that actually engage players — not generic, flat narratives.

3.1 Fiction Genres

Genre	Subgenres	Core Engagement Mechanic
Fantasy	High/Epic, Dark, Urban, Sword & Sorcery, Mythological, Fairy Tale Retelling, Steampunk, Progression/LitRPG	Wonder + consistent internal rules. Magic must have costs and limitations. Flawed heroes, nuanced villains.
Sci-Fi	Space Opera, Cyberpunk, Hard Sci-Fi, Post-Apocalyptic, Dystopian, Solarpunk, Military Sci-Fi, Time Travel, First Contact	"What if?" extrapolation. One speculative premise explored rigorously. Human stakes inside the spectacle.
Horror	Gothic, Cosmic/Lovecraftian, Slasher/Survival, Psychological, Paranormal, Body Horror, Folk Horror	Dread over shock. Keep threats ambiguous. Start mundane, turn sinister. Fear of the unknown > explicit description.
Mystery	Whodunit, Cozy, Hardboiled/Noir, Police Procedural, Locked Room, True Crime-style	The puzzle the player co-solves. 4–7 suspects with motives. Real clues + red herrings. Design from villain's POV first.
Thriller	Espionage, Legal, Medical, Political, Techno-Thriller, Psychological	Ticking-clock pressure + shifting trust. Player knows the danger, must outmaneuver it. Constant escalation.
Romance	Contemporary, Historical, Paranormal, Enemies-to-Lovers, Slow Burn, Romantic Suspense	Character transformation through love. Push-pull dynamic. Emotional wounds as barriers. Three arcs: each lead + relationship.
Western	Classic Frontier, Weird West, Revisionist, Outlaw	Moral codes under pressure. Vast, hostile landscape as character. Justice vs survival tension.
Historical Fiction	Ancient World, Medieval, Renaissance, Colonial, Victorian, WWI/WWII, Cold War	Verisimilitude + dramatic license. Real events as backdrop for personal stakes. Period-accurate sensory detail.
Literary Fiction	Coming-of-Age, Family Saga, Magical Realism, Existential, Epistolary	Internal transformation. Prose quality and thematic depth over plot. Character interiority drives engagement.
Comedy/Satire	Parody, Dark Comedy, Absurdist, Workplace Comedy, Romantic Comedy	Subverted expectations. Comedic timing in text pacing. Social commentary through humor.

3.2 Non-Fiction Inspired Genres

Genre	Subgenres	Core Engagement Mechanic

Survival/Adventure	Wilderness, Lost at Sea, Expedition, Mountaineering, Disaster	Resource tracking, environmental escalation, physical consequences compound over time
True Crime (Fictional)	Serial Killer Investigation, Cold Case, Organized Crime, Heist	Evidence-based progression, moral ambiguity of investigator, victim humanization
War & Military	Historical Battle, Spec-Ops, Resistance/Guerrilla, Naval, Siege	Chain of command tension, impossible moral choices, camaraderie under fire
Political Drama	Election Campaign, Diplomatic Crisis, Coup d'État, Courtroom, Legislative	Competing factions, public vs private face, cost of compromise
Business/Corporate	Hostile Takeover, Startup, Corporate Espionage, Wall Street, Whistleblower	Power dynamics, information asymmetry, the deal as battle, moral cost of ambition
Journalism	Investigative Reporter, War Correspondent, Whistleblower, Undercover	Truth vs safety tension, source protection, institutional resistance
Sports	Underdog Story, Coaching Drama, Olympic Journey, Boxing/MMA, Racing	Training progression, mentor relationship, cost of obsession, the big moment
Exploration	Archaeology, Deep Sea, Space (realistic), Arctic/Antarctic, Caving	Discovery-driven wonder, environmental danger, isolation, the unknown
Biographical (Fictional)	Rise to Power, Artist's Journey, Criminal Empire, Scientific Discovery	Character transformation through ambition, historical authenticity, defining choices

3.3 Genre Crossover System

Players can optionally select a crossover tag to blend two genres, producing hybrid experiences. The AI merges the craft rules of both genres, using the primary genre for world/setting and the secondary for narrative tension mechanics. Examples: Sci-Fi + Horror = Alien survival; Romance + Thriller = Romantic suspense; Historical + Mystery = Victorian whodunit; Business + Thriller = Corporate espionage; Western + Horror = Weird West; Fantasy + Romance = Romantic fantasy.

4. Genre Craft Rules (AI Agent Instructions)

These rules are injected into the Story Director and Scene Writer system prompts based on the selected genre. They encode what master storytellers and screenwriters know about making each genre actually work.

4.1 Fantasy Craft Rules

- Generate a consistent magic system with defined rules, costs, and limitations. Magic that solves everything destroys tension.
- Protagonists must have meaningful flaws — no generic heroes. Internal conflict is as important as external.
- Villains need empathy-inducing motivations, not cartoon evil. The best antagonists believe they're right.
- Deliver worldbuilding through action and dialogue, never info-dumps. The reader learns alongside the protagonist.
- Start scope small (a village, a personal quest), reveal the larger stakes gradually. Scope escalation creates momentum.
- Subgenre specifics: Dark Fantasy needs moral ambiguity and cost; LitRPG needs visible progression systems; Steampunk needs inventive tech flavor woven into culture.

4.2 Horror Craft Rules

- Prioritize dread over shock. Dread is sustained, escalating tension — a bomb you can't find. Jump scares are firecrackers.
- Keep threats ambiguous as long as possible. The moment you fully describe the monster, it becomes a problem to solve, not a source of fear.
- Start in mundane, relatable settings that turn sinister. The more familiar the setting, the scarier the corruption.
- Use sensory detail across all five senses to build atmosphere — sounds, smells, temperature, texture, not just visuals.
- Deploy false safety beats — moments where everything seems fine that lull the player before the next escalation.
- Gothic: atmospheric description + psychological tension. Psychological: seed subtle inconsistencies so the player questions reality. Survival: track resources and compound physical consequences. Cosmic: emphasize insignificance and incomprehensibility.

4.3 Mystery/Noir Craft Rules

- Design the story from the villain's perspective first — establish what happened and why, then layer the player's investigation on top.
- Maintain 4–7 suspects, each with a clear motive, means, and opportunity — plus an unrelated secret that makes them suspicious for the wrong reasons.
- Plant real clues AND red herrings. Track which the player has found. The player should be able to solve it one scene before the reveal if they're paying attention.
- Clues should be seen but misinterpreted, or seen but overlooked by the narrative — the player is led astray by reasonable assumptions.

- Cozy: character charm, off-page violence, warm tone. Noir: moral ambiguity, cynical realism, atmospheric prose. Procedural: methodical evidence analysis, institutional politics.

4.4 Thriller Craft Rules

- The player often knows the danger early — the tension comes from trying to prevent or survive it, not discovering it.
- Use ticking-clock pressure: every scene should feel like time is running out.
- Build trust with NPCs specifically so it can be shattered — betrayal architecture requires investment before the payoff.
- Constantly escalate stakes. Each scene raises the danger level. No plateaus.
- Espionage: layers of deception, loyalty tests. Legal: procedural tension, moral compromise. Psychological: unreliable perspectives, gaslighting.

4.5 Romance Craft Rules

- Each lead must have a clear emotional wound or limiting belief that makes love difficult. Without internal barriers, conflict feels artificial.
- The relationship is its own arc — a third character. Three steps forward, two steps back. Draw characters together, then force them apart, repeatedly.
- Chemistry is built through contrast, banter, vulnerability, and subtext — not declarations. Show through actions and reactions, not telling.
- Avoid insta-love. Characters who agree on everything and have no conflict produce boring stories. Dynamic relationships challenge both characters.
- The climactic resolution must feel earned through mutual growth and transformation, not convenient plot resolution.

4.6 Sci-Fi Craft Rules

- Establish one speculative premise and explore its consequences rigorously. Internal consistency is paramount.
- Technology/concepts are the setting, not the story. Character drives engagement. Human stakes inside the spectacle.
- The paradigm-shift moment — when the player's understanding of the world fundamentally changes — is the genre's signature payoff.
- Hard Sci-Fi: plausible science, real physics constraints. Cyberpunk: class/power commentary, sensory overload. Post-apocalyptic: resource tension, moral impossible choices.

4.7 Non-Fiction Inspired Craft Rules

- Verisimilitude is the core requirement. These genres must feel like they could really happen, even in a fictional frame.
- Survival: resource tracking and environmental escalation with compounding physical consequences (injuries don't disappear).
- War: chain of command tension, impossible moral choices, the weight of responsibility, camaraderie that deepens loss.
- Business/Corporate: information asymmetry as the weapon, the deal as a battlefield, moral cost of ambition.

- Political: every character has a public face and a private agenda. Compromise always has a cost.
- Sports: visible progression through training, the mentor relationship, the physical and psychological cost of obsession.

5. Plot Twist Framework

When plot twists are enabled, the AI follows a structured framework based on master storytelling craft from Sanderson, screenwriting (Final Draft, MasterClass), and narrative design. Twists are never random — they are foreshadowed, genre-appropriate, and earn their impact.

5.1 Universal Principles

- A twist must feel both surprising AND inevitable in hindsight. If the player can't look back and see the seeds, it's cheap.
- The Story Director maintains a twist_seeds[] array — planting 2–3 foreshadowing details across scenes before triggering any twist.
- Every twist must reference at least one prior seed. No twists from nowhere.
- A POV character can omit details or misinterpret clues, but the narrative must never directly lie to the player via inner thought.
- Twists should give MORE than expected, not just shock — they should recontextualize what came before and raise stakes.

5.2 Sanderson's Seven Twist Types (Adapted)

Twist Type	Description	Best For
Big Plot / Little Plot	A subplot from early in the story becomes the main plot. The scope is much larger than anticipated.	Fantasy, Sci-Fi, Thriller
Plausible Deniability	Threaten to break a promise to the player, make them worry, then deliver — but in an unexpected way.	Romance, Mystery, Horror
Failure	The protagonist actually fails at their quest. Subverts genre expectations that the hero always wins.	Fantasy, Thriller, War
Smoke & Mirror	Foreshadow one thing while actually setting up something bigger. Double-duty foreshadowing.	Sci-Fi, Mystery, Horror
Look at the Rabbit	Introduce a plot element early, make the player forget about it, then erupt it back into the story.	All genres
Escalation	Not a reversal but a massive increase in scope, danger, or stakes that changes everything.	Thriller, Fantasy, War
Identity Reveal	A character is not who they appeared to be — ally was enemy, enemy was ally, hidden relationship.	Mystery, Noir, Romance, Horror

5.3 Genre-Specific Twist Rules

Fantasy

- Preferred: Scope escalation, prophecy reinterpretation, hidden identity, magic system reveals that recontextualize everything
- Plant seeds in: worldbuilding details, NPC backstories, magic system edge cases

Horror

- Preferred: False safety inversions, the real monster reveal, protagonist-as-threat, "we were the ghosts" inversion
- Plant seeds in: environmental inconsistencies, NPC behavior patterns, sensory details that don't quite fit

Mystery/Noir

- Preferred: The double-cross, cascading revelations (each answer opens new questions), unreliable witness, suspect pool inversion
- Plant seeds in: dialogue inconsistencies, alibi details, physical evidence placement, NPC secrets

Thriller

- Preferred: Betrayal by trusted ally, ticking-clock reversal (time pressure was manufactured), protagonist's own complicity
- Layer small twists into larger ones — small reveal, then immediately a bigger one while the player is still processing

Romance

- Preferred: Secret identity/hidden past, misunderstanding recontextualization, the unexpected pairing, sacrifice twist
- High-risk: subverting the HEA (Happy Ever After) expectation. Only do this if the alternative is more satisfying.

Sci-Fi

- Preferred: Paradigm shift (the reality assumed was wrong), time-loop reveals, technology-as-betrayal, "it was Earth all along"
- Plant seeds in: worldbuilding flavor text, technical details that seem like exposition, background environmental details

5.4 Twist Quality Checklist (AI Self-Validation)

Before the Story Director triggers a twist, it must validate against this checklist:

- Was it foreshadowed with at least 2 planted seeds in prior scenes? If no → delay twist, plant more seeds.
- Does it change the meaning of previous scenes when the player looks back? If no → it's not a real twist.
- Does it raise stakes or deepen the story? If no → it's just shock value.
- Is it consistent with established character motivations and world rules? If no → it will feel cheap.
- Would a player who was paying close attention have had a chance to guess it one scene before? If no → more foreshadowing needed.

6. AI Agent Architecture

6.1 Multi-Provider, Model-Agnostic Strategy

Loreforge uses a provider abstraction layer that routes each agent task to the optimal model based on cost, quality, and latency requirements. No vendor lock-in. The routing table is configurable per deployment.

Agent	Task	Free Tier Model	Premium Model	Cost/Scene
Story Director	Plot arc, tension, twist seeds, pacing	DeepSeek V3.2 ($0.28/$0.42)	Claude Haiku 4.5 ($1/$5)	~$0.001–$0.003
Scene Writer	Narrative prose generation (STREAMING)	DeepSeek V3.2 or Gemini Flash	Claude Haiku 4.5 or GPT-5 Mini	~$0.001–$0.005
Choice Generator	2–4 contextual player choices	GPT-5 Nano ($0.05/$0.40)	GPT-5 Mini ($0.25/$2)	~$0.0003
Skill Check Arbiter	Set DCs, resolve rolls, flavor text	Rule-based (no LLM) / Gemini Flash-Lite	GPT-5 Nano	~$0.0001
Reference Agent	Analyze favorite stories → taste profile (one-time)	DeepSeek V3.2	Claude Haiku 4.5	~$0.003 (once)
Consistency Auditor	Validate scene against world state for contradictions	Rule-based + GPT-5 Nano	Claude Haiku 4.5	~$0.0002
World State Manager	Track inventory, relationships, stats, flags	Deterministic logic (no LLM)	Deterministic logic (no LLM)	$0
Visual Director	Select background, sprites, music per scene	Classifier / tag matching	Classifier / tag matching	$0

Estimated cost per scene (free tier): $0.002–$0.005. At 100 scenes per player per month, monthly API cost per free user: $0.20–$0.50. Premium users get higher-quality prose for ~2–3x the cost, funded by subscription revenue.

Cache strategy: System prompts and World Bible content use prompt caching (90% savings on repeated context). DeepSeek and Anthropic both support this natively.

6.2 Streaming Architecture

The Scene Writer uses server-sent events (SSE) streaming so narrative text appears word-by-word as it's generated. This eliminates the 2–5 second wait for full generation and creates an immersive "story unfolding" effect.

- Scene Writer output streams directly to the UI text box with a blinking cursor
- While the player reads streaming text, the Choice Generator runs in parallel (non-streaming) so choices are ready the moment the narrative completes
- Loading indicators show which agents are active: "Story Director thinking..." → "Writing scene..." → "Generating choices..."

6.3 Prompt Architecture

Each AI call uses a structured prompt composed of layered context:

1. System Prompt: Agent role, output format (JSON), constraints, genre craft rules from Section 4
2. World Bible: Compiled from Session Zero — genre rules, tone guide, protagonist profile, taste profile from favorite stories
3. Genre Craft Rules: Injected from Section 4 based on selected genre/subgenre
4. Twist State: Current twist_seeds[] array, twist_count, scenes_since_last_twist (if twists enabled)
5. Narrative State: Current act/chapter, active plot threads, tension curve, unresolved conflicts
6. Scene Context: Last 3–5 scenes as compressed summary (rolling compression every 5 scenes)
7. World State Snapshot: Stats, inventory, relationships, location, time, flags as structured JSON
8. Player Action: The choice or free-text input the player just made

6.4 Context Window Management

To avoid context limits, the system uses rolling summary compression. A dedicated summarization call compresses older scenes into structured summaries every 5 scenes. The full context payload targets 4,000–6,000 tokens per generation call. For long stories, critical plot points, twist seeds, and relationship states are preserved even when scene prose is compressed.

7. Game Mechanics (RPG Mode)

These mechanics apply when the player selects RPG Adventure mode. In Pure Story mode, all game mechanics are hidden and the experience is purely narrative with choices.

7.1 Zork-Style Exploration Engine

The game includes a full text adventure parser supporting classic commands alongside the visual novel choice interface:

- Movement: go [direction], north/south/east/west/up/down
- Interaction: take, drop, use [item] on [target], examine, search, open
- Social: talk to [NPC], attack [NPC]
- Meta: look, inventory, score, help
- Free-text fallback: anything the parser doesn't recognize goes to the AI for creative interpretation

The room graph system tracks interconnected rooms with exits, items, darkness states, and locked passages. Dark rooms require a light source item. Locked exits require specific key items. The AI can dynamically generate new rooms and items that integrate into the existing graph.

7.2 Stat & Skill Check System

Stat	Governs	Example Usage
Strength	Physical power, intimidation	Break down a door, win a fight
Agility	Speed, stealth, reflexes	Dodge an attack, pick a lock
Intelligence	Logic, knowledge, puzzles	Decipher runes, outsmart a con artist
Charisma	Persuasion, deception, leadership	Negotiate a treaty, convince an NPC
Perception	Awareness, tracking, intuition	Notice a hidden trap, read body language
Willpower	Mental fortitude, resistance	Resist manipulation, endure psychological pressure

Skill checks: d20 + stat modifier vs Difficulty Class (DC) set by the AI. Animated dice roll on screen. Nat 20 = critical success (special narrative outcome). Nat 1 = critical failure. The Lite RPG preset uses 3 stats (Mind/Body/Social) with simple pass/fail instead of the full d20 system.

7.3 Inventory, Relationships & Scoring

- Contextual inventory: items are genre-appropriate (potions in fantasy, medkits in sci-fi, evidence in mystery)
- AI tracks inventory state and references items in narrative and choices. Key items can unlock unique story paths.
- NPC affinity tracked on a −10 to +10 scale. Choices shift affinity, unlocking loyalty quests, betrayals, or romance arcs.
- Scoring system (Zork-style): points awarded for exploration, puzzle solving, item discovery, and story milestones.

8. Technical Architecture

8.1 Tech Stack

Layer	Technology	Notes
Game Engine	Flutter + Flame 1.x	Cross-platform rendering, sprite system, audio
UI Framework	Flutter (Material/Cupertino)	Dialogue boxes, menus, HUD, Session Zero wizard
State Management	Riverpod + Freezed	Immutable world state, reactive UI updates
Local Database	Drift (SQLite)	Save files, player profiles, asset cache
AI Client	Provider abstraction layer (Dart)	Routes to DeepSeek/Gemini/OpenAI/Anthropic APIs with SSE streaming
AI Local Runtime	llama.cpp via dart:ffi	On-device inference for offline fallback (quantized 8B models)
Asset Pipeline	Sprite sheets + Spine 2D	Character animations, expressions, genre variants
Audio	flame_audio + audioplayers	Adaptive BGM, ambient SFX, genre-matched
Backend	Supabase / Firebase	Cloud saves, analytics, user accounts, subscription management
CI/CD	GitHub Actions + Fastlane	Automated builds for iOS, Android, desktop

8.2 Provider Abstraction Layer

The AI client implements a unified interface that abstracts provider differences. Each provider adapter handles authentication, request formatting, response parsing, streaming (SSE), and error handling. The routing table maps (agent_type, user_tier) to (provider, model). Switching providers or adding new ones requires only a new adapter — no changes to game logic.

- Supported providers: Anthropic (Claude), OpenAI (GPT), Google (Gemini), DeepSeek
- Streaming support: All providers via SSE. Unified onChunk callback interface.
- Fallback chain: If primary provider fails, auto-retry with secondary provider before showing error.
- Cost tracking: Per-request cost logging for analytics and budget management per user tier.

9. UI/UX Design

9.1 Screen Flow

1. Splash / Title Screen → Main Menu
2. Main Menu: New Adventure | Continue | Gallery | Settings
3. New Adventure → Session Zero Wizard (8–10 screens, card-based with visual/audio previews)
4. Favorite Stories input (free-text, with example suggestions)
5. RPG Depth configuration (presets + optional granular toggles)
6. Session Zero Complete → Genre-themed animated intro
7. Gameplay Screen: Background + Sprites + Streaming Text Box + Choice Panel
8. In-Game Overlays: Journal, Inventory, Character Sheet, Map, Settings
9. Story End → Epilogue + Stats Summary + Share + Play Again

9.2 Streaming Text UX

The gameplay text box shows a blinking amber cursor as narrative text streams in word-by-word. This creates the feeling of a story being told in real time rather than waiting for a block of text. The player can scroll up at any time to re-read. Choices slide up from the bottom only after the narrative stream completes. Agent status indicators show pipeline progress: small animated dots with labels like "Writing scene..." or "Generating choices..."

10. Monetization Strategy

Tier	Price	Features
Free	$0	3 adventures/week, 2 genres (Fantasy + Mystery), short stories only, DeepSeek/Gemini models, ads between sessions
Adventurer	$4.99/mo	Unlimited adventures, all genres + subgenres, medium + long stories, no ads, cloud saves, favorite stories input
Loremaster	$9.99/mo	Everything + premium AI models (Claude/GPT-5 Mini), full RPG depth, offline mode, party system, exclusive sprite packs, priority API
Genre Packs	$1.99–$4.99	Individual genre expansion packs (new sprites, backgrounds, music per genre), cosmetic UI themes

API cost management: Free tier uses cheapest models (~$0.003/scene). At ~$0.30/month per active free user, the free tier is sustainable with modest ad revenue. Adventurer tier margins are ~80% after API costs. Loremaster tier uses premium models but subscription covers the ~$1–2/month API cost with healthy margin.

11. Development Roadmap

Phase	Timeline	Deliverables
Phase 0: Prototype	Weeks 1–4	CLI story loop, multi-provider AI pipeline, streaming proof of concept, genre craft rule validation
Phase 1: Core Engine	Weeks 5–12	Flutter Flame scaffold, gameplay screen, streaming text renderer, Zork parser, world state engine, provider abstraction layer
Phase 2: Session Zero	Weeks 13–18	Full wizard with all 10 steps, favorite stories analyzer, RPG depth presets + toggles, genre/subgenre selector with 30+ options
Phase 3: Genre & Twist System	Weeks 19–22	Genre craft rules injected into prompts, twist framework with seed tracking, genre-specific twist validation
Phase 4: RPG Layer	Weeks 23–26	Full stat system, d20 skill checks with animations, inventory, relationship tracker, scoring, companion system
Phase 5: Visual Polish	Weeks 27–32	Character sprite system (per-genre), parallax backgrounds, adaptive music, tag-based asset matching, VFX
Phase 6: Offline Mode	Weeks 33–36	Local model integration (llama.cpp), model download manager, quality toggle, hybrid cloud/local switching
Phase 7: Platform & QA	Weeks 37–40	Cloud saves, user accounts, cross-platform testing, performance optimization, beta testing, monetization integration
Phase 8: Launch	Weeks 41–42	App Store / Play Store / Steam submission, marketing, analytics setup
Phase 9: Post-Launch	Ongoing	New genre packs, multiplayer co-op, AI image generation, community features, additional languages

Total estimated timeline to MVP (Phase 0–3, core loop + Session Zero + genre system): 22 weeks (~5.5 months) with 3–5 developers. Solo developer timeline: 10–12 months targeting Phases 0–2 as a playable vertical slice with 2 genres.

12. Technical Risks & Mitigations

Risk	Severity	Mitigation
AI generates flat/boring genre-specific content	Critical	Genre craft rules (Section 4) injected into every prompt. Consistency Auditor validates engagement markers. Taste profile personalizes output.
Plot twists feel cheap or random	High	Twist framework (Section 5) requires seed planting + self-validation checklist. No twist without 2+ prior seeds.
Provider API outage or rate limiting	High	Multi-provider fallback chain. If primary fails, auto-switch to secondary. Local model as last resort.
AI generates incoherent or contradictory narrative	High	Consistency Auditor validates every scene against world state JSON. Auto-retry on contradiction.
API costs spiral with free users	High	Cheapest models for free tier (DeepSeek/Gemini Flash-Lite). Rate limiting. Prompt caching saves 90%.
Streaming latency varies across providers	Medium	Pre-generate next scene in background. Show loading animation styled as story element.
Context window overflow on long stories	Medium	Rolling summary compression every 5 scenes. Hard cap at 6K tokens per call.
Player writes adversarial/NSFW free-text input	High	Content moderation layer before AI ingestion. Safety system prompt guardrails.
Favorite stories analysis produces weak taste profile	Medium	Fallback to genre defaults if reference works are obscure. Allow manual taste profile editing.

13. Minimum Viable Product (MVP) Scope

In Scope

- Session Zero wizard with language, mode, setup method (prompt or questions), genre, tone, favorite stories, twists toggle
- 4 fully-supported genres: Fantasy, Horror, Mystery, Sci-Fi (with subgenres and craft rules)
- Core gameplay loop with streaming AI narrative → choices → outcome → next scene
- RPG mode with Story+ and Lite RPG presets (full d20 system post-MVP)
- Pure Story mode (no game mechanics)
- Multi-provider AI pipeline (DeepSeek + Gemini Flash for free tier)
- Favorite stories taste profile system
- Plot twist framework with seed tracking
- Zork-style text parser (RPG mode)
- 9 languages
- Mobile only (iOS + Android)

Out of Scope (Post-MVP)

- Desktop builds, offline/local model support
- Full d20 stat system, companion/party system, crafting
- AI image generation for custom scene art
- Multiplayer co-op storytelling
- Cloud saves and user accounts
- Remaining 20+ genres (launched as genre packs)
- Monetization (free during beta)

E N D O F D O C U M E N T